

Introduction to ADCs

Figure 10.5-1 shows a block diagram of a general ADC. A prefilter called an *antialiasing filter* is necessary to avoid the aliasing of higher frequency signals back into the baseband of the ADC. Often, the antialiasing filter is implemented by the bandlimiting characteristics of the ADC itself. The antialiasing filter is followed by a sample-and-hold circuit that maintains the input analog signal to the ADC constant during the time this signal is converted to an equivalent output digital code. This period of time is called the *conversion time* of the ADC. The conversion is accomplished by a quantization step. The nature of a quantizer is to segment the reference into subranges. Typically, there are 2^N subranges, where N is the number of bits of the digital output code. The quantization step finds the subrange that corresponds to the sampled analog input. Knowing this subrange allows the digital processor to encode the corresponding digital bits. Thus, within the conversion time, a sampled analog input signal is converted to an equivalent digital output code.

The frequency response of the ADC of Fig. 10.5-1 is important to understand. Let us assume that the analog input signal has the frequency response shown in Fig. 10.5-2(a). Furthermore, assume that the frequency, f_B , is the highest frequency of interest of the analog input signal. When the analog input signal is sampled at a frequency of f_s , the frequency response shown in Fig. 10.5-2(b) results. The spectrum of the input signal is aliased at the sampling frequency and each of its harmonics. If the bandwidth of the signal, f_B , is increased above $0.5f_s$, the spectra begin to overlap as shown in Fig. 10.5-2(c). At this point it is impossible to recover the original signal. This concept is formalized in the *Nyquist frequency or rate*, which states that the sampling frequency must be at least twice the bandwidth of the signal in order for the signal to be recovered from the samples. Consequently, it is necessary to apply the prefilter of

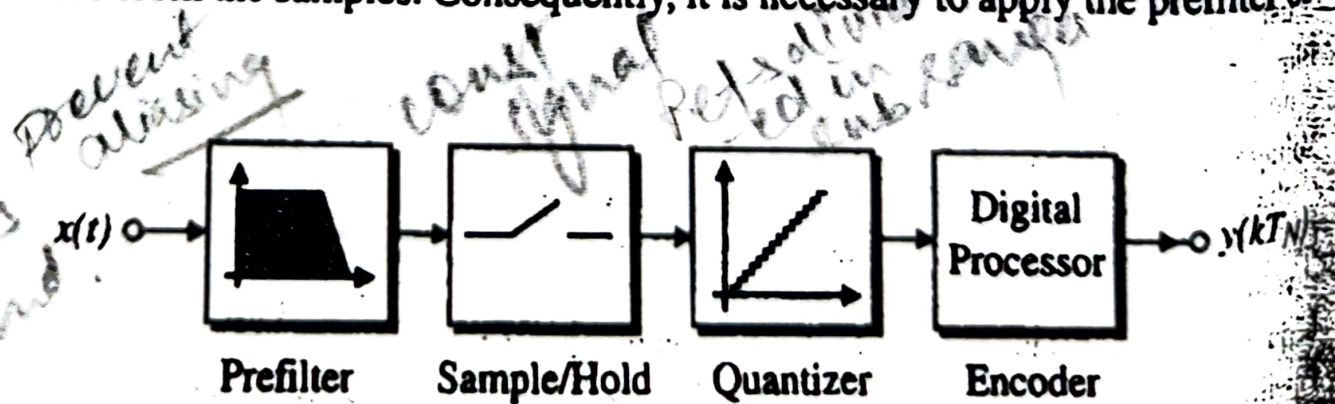


Figure 10.5-1 General block diagram for an ADC.

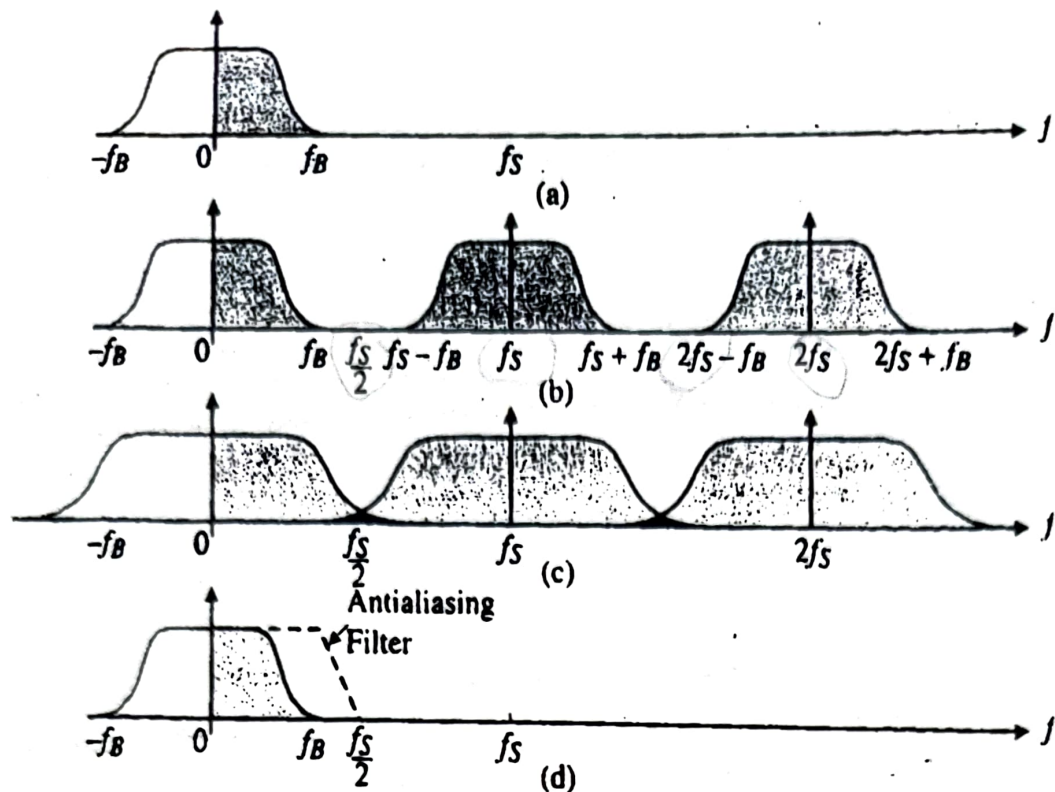


Figure 10.5-2 (a) Continuous time frequency response of the analog input signal. (b) Sampled-data equivalent frequency response. (c) Case where f_B is larger than $0.5f_s$, causing aliasing. (d) Use of an antialiasing filter to avoid aliasing.

Fig. 10.5-1 to eliminate signals in the incoming analog input that are above $0.5f_s$. This is shown in Fig. 10.5-2(d). The overlapping of the folded spectra also will occur if the bandwidth of the analog input signal remains fixed but the sampling frequency decreases below $2f_B$. Even if f_B is less than $0.5f_s$ as in Fig. 10.5-2(b), as we have seen in the previous chapter the antialiasing filter is necessary to eliminate the aliasing of signals in the upper passbands into the baseband which is from 0 to f_B .

In order to maximize the input bandwidth of the ADC, one desires to make f_B as close to $0.5f_s$ as possible. Unfortunately, this requires a very sharp cutoff for the prefilter or antialiasing filter, which make this filter difficult and complex to implement. The types of ADCs that operate in this manner are called *Nyquist analog-to-digital converters*. Later we will examine ADCs that have f_B much less than $0.5f_s$. These ADCs are called *oversampling analog-to-digital converters*. Table 10.5-1 gives the classification of various types of ADCs that will be discussed in this chapter.

TABLE 10.5-1 Classification of ADC Architectures

Conversion Rate	Nyquist ADCs	Oversampled ADCs
Slow	Integrating (serial)	Very high resolution > 14 bits
Medium	Successive approximation 1-bit Pipeline Algorithmic	Moderate resolution > 10 bits
Fast	Flash Multiple-bit pipeline Folding and interpolating	Low resolution > 6 bits

Static Characterization of ADCs

The input of an ADC is an analog signal, typically an analog voltage, and the output is a digital code. The analog input can have any value between 0 and V_{REF} while the digital code is restricted to fixed or discrete amplitudes. Popular digital codes used for ADCs are shown in Table 10.5-2 and include binary, thermometer, Gray, and two's complement. The most widely used digital code is the binary code. Some codes have advantages over others that make them attractive. For example, the Gray and thermometer codes only change 1-bit from one code to the next.

The static characterization of ADCs is based on the input-output characteristic shown in Fig. 10.5-3 for a 3-bit ADC. In this particular characteristic, the input has been shifted so that the ideal step changes occur at analog input values of $0.5LSB(2i - 1)$, where i varies from 1 to N for an N -bit ADC.

Beneath the input-output characteristic of Fig. 10.5-3 is a plot of the quantization noise as a function of the input. The quantization noise is a plot of the difference between the infinite resolution characteristic and the ideal 3-bit characteristic as a function of the input voltage. The ideal ADC characteristic will have a quantization noise that lies between $\pm 0.5LSB$.

The definitions for *dynamic range*, the *signal-to-noise ratio (SNR)*, and the *effective number of bits (ENOB)* of the ADC are the same as those given in Section 10.1 for the DAC. These quantities were referenced to the analog variable and in the case of the ADC are referenced to the digital output word.

The *resolution* of the ADC is the smallest analog change that can be distinguished by an ADC. Resolution may be expressed in percent of full scale (FS) but is typically given in the number of bits, N , where the converter has 2^N possible output states.

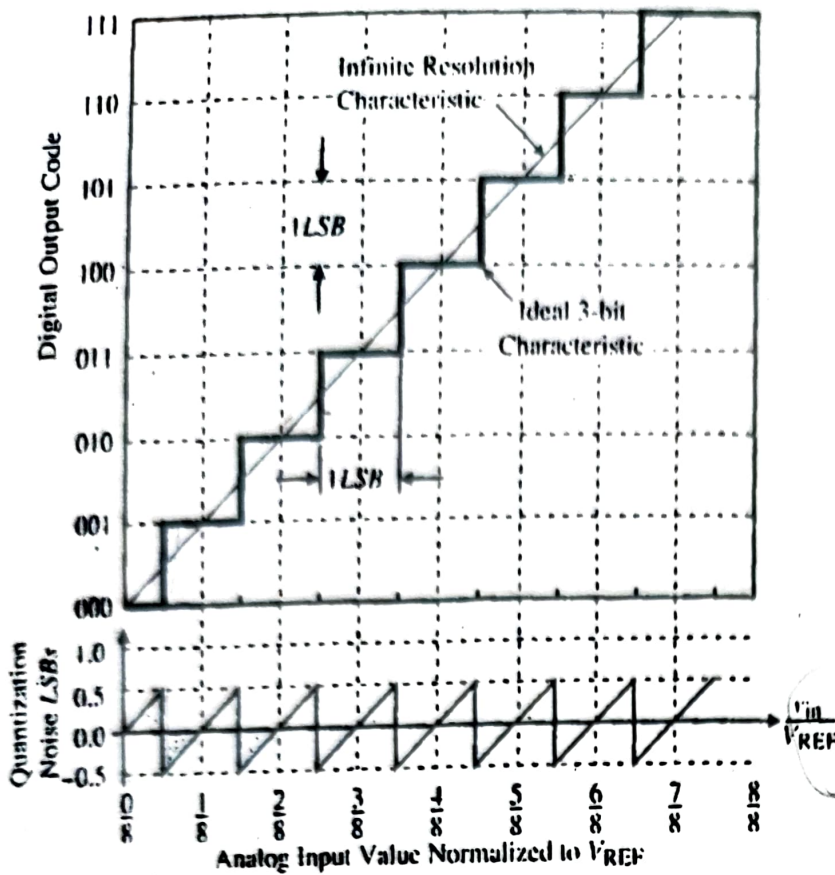
The primary characteristics that define the static performance of converters are *offset error*, *gain error*, *integral nonlinearity (INL)*, and *differential nonlinearity (DNL)*. For an ADC with offset, let us shift the infinite resolution characteristic line horizontally until the quantization noise is symmetrical when referenced to this line (here we are assuming that other errors such as gain and nonlinearity are not dominant or have been removed from the characteristic). The horizontal difference between this line and the infinite resolution characteristic that passes through the origin is *offset error*. Offset error is illustrated in Fig. 10.5-4(a).

Gain error is a difference between the actual characteristic, and the infinite resolution characteristic, which is proportional to the magnitude of the input voltage. The gain error can be thought of as a change in the slope of the infinite resolution line above or below a value of 1.

TABLE 10.5-2 Digital Output Codes Used for ADCs

Decimal	Binary	Thermometer	Gray	Two's Complement
0	000	000000	000	000
1	001	000001	001	111
2	010	000011	011	110
3	011	000111	010	101
4	100	001111	110	100
5	101	011111	111	011
6	110	011111	101	010
7	111	111111	100	001

Figure 10.5-3 Ideal input-output characteristics of a 3-bit ADC.



Gain error is illustrated in Fig. 10.5-4(b). Similar to the DAC, gain error can be measured as the horizontal difference in *LSBs* between actual and ideal finite resolution characteristics at highest digital code, i.e., between 110 and 111 on Fig. 10.5-4(b). In this example, it is assumed that all other errors such as offset and nonlinearity are not present.

The definition for integral nonlinearity (*INL*) of the ADC is the maximum difference between the actual finite resolution characteristic and the ideal finite resolution characteristic measured vertically in percent or *LSBs*. With this definition, we find that only integer values are permitted because the digital output codes correspond to discrete amplitudes. This is not a problem as the resolution increases and the *LSB* becomes small. In addition, when measuring the *INL* if the measurement equipment is sufficiently accurate, it is able to resolve the *INL* to less than a *LSB*.

Small LSB → high Resolution

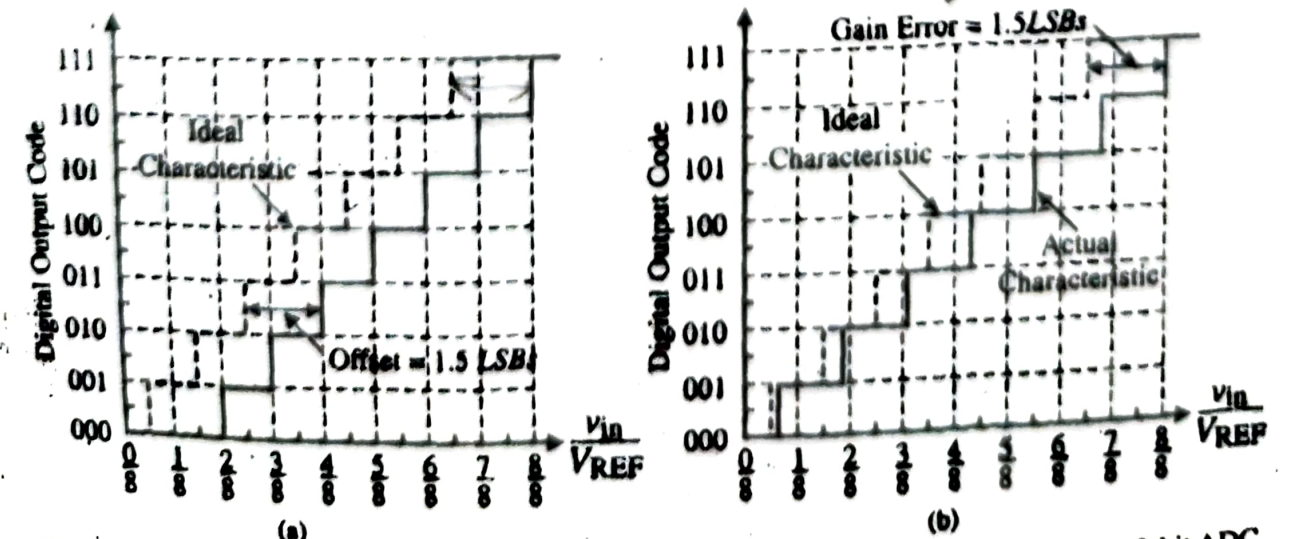


Figure 10.5-4 (a) Example of offset error for a 3-bit ADC. (b) Example of gain error for a 3-bit ADC.

Differential nonlinearity (DNL) of the ADC is defined as a measure of the separation between adjacent codes measured at each vertical step in percent or *LSBs*. The differential nonlinearity of an ADC can be written as

$$DNL = (D_{ex} - 1) \text{LSBs} \quad (10.5-1)$$

where D_{ex} is the size of the actual vertical step in *LSBs*. Figure 10.5-5 shows the integral and differential nonlinearity for a 3-bit ADC referenced to the digital output code. We see that the largest and smallest values of *INL* are $+1\text{LSB}$ and -1LSB , respectively. The largest and smallest values of *DNL* are $+1\text{LSB}$ and 0LSB , respectively. As compared to the DAC, a *DNL* of $-\text{LSB}$, which resulted from the case where a step should have occurred, does not happen for the ADC. For example, at an input voltage of $\frac{3}{16}$ on Fig. 10.5-5, the fact that a vertical jump does not occur cannot be considered as a *DNL* of -1LSB .

Nonmonotonicity in an ADC occurs when a vertical jump is negative. Nonmonotonicity can only be detected by *DNL*. Because output is limited to digital codes, all jumps are integers. Normally, the vertical jump is 1LSB . If the jump is 2LSBs or greater, missing output codes may occur. If the vertical jump is less than 0LSB , then the ADC is not monotonic. Figure 10.5-6 shows a 3-bit ADC characteristic that is not monotonic. Nonmonotonicity generally occurs when the *MSB* does not have sufficient accuracy. The change from $0111 \dots$ to $10000 \dots$ is the most difficult because the *MSB* must have the accuracy of $\pm 0.5\text{LSB}$ or excessive *DNL* will occur.

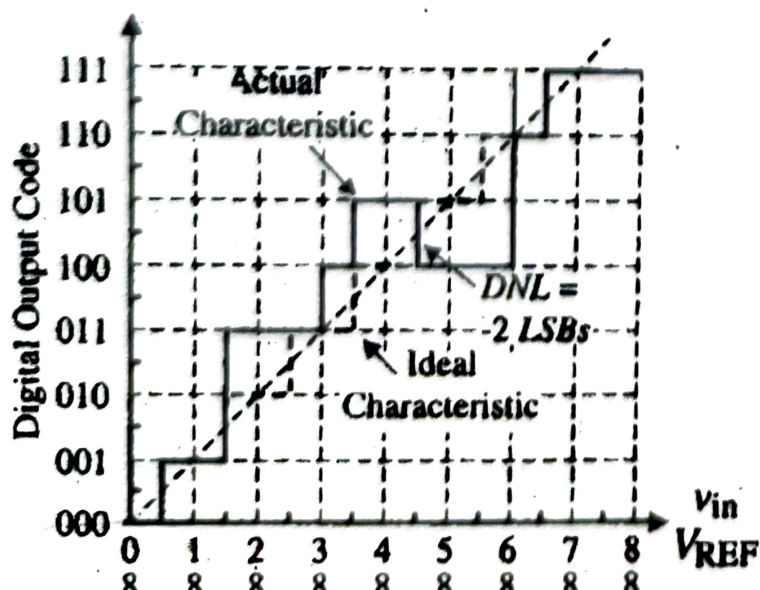


Figure 10.5-6 Example of nonmonotonic 3-bit ADC.

PRODUCTION AND CHARACTERIZATION OF DIGITAL-ANALOG CONVERTERS

This section examines the digital-to-analog conversion aspect of this important interface. Figure 10.1-1 illustrates how DACs are used in data systems [1]. The input to a DAC is a digital word consisting of parallel binary signals that are generated from a digital signal-processing system. These parallel binary signals are converted to an equivalent analog signal by scaling a reference. The analog output signal may be filtered and/or amplified before being applied to an analog signal-processing system. While the output may be voltage or current, most DACs have a voltage output.

A DAC with a voltage output can be characterized by the block diagram in Fig. 10.1-2(a). We see that it consists of a digital word of N -bits ($b_0, b_1, b_2, \dots, b_{N-2}, b_{N-1}$) and a reference voltage V_{REF} . b_0 is called the *most significant bit*, MSB, and b_{N-1} is called the *least significant bit*, LSB. The voltage output v_{OUT} can be expressed as

$$v_{OUT} = KV_{REF}D \quad (10.1-1)$$

where K is a scaling factor and the digital word D is given as

$$D = \frac{b_0}{2^1} + \frac{b_1}{2^2} + \frac{b_2}{2^3} + \dots + \frac{b_{N-1}}{2^N} \quad (10.1-2)$$

N is the total number of bits of the digital word and b_{i-1} is the i th-bit coefficient and is either 0 or 1. Therefore, the output of a DAC can be expressed by combining Eqs. (10.1-1) and (10.1-2) to get

$$v_{OUT} = KV_{REF} \left(\frac{b_0}{2^1} + \frac{b_1}{2^2} + \frac{b_2}{2^3} + \dots + \frac{b_{N-1}}{2^N} \right) \quad (10.1-3)$$

or

$$v_{OUT} = KV_{REF} [b_0 2^{-1} + b_1 2^{-2} + b_2 2^{-3} + \dots + b_{N-1} 2^{-N}] \quad (10.1-4)$$

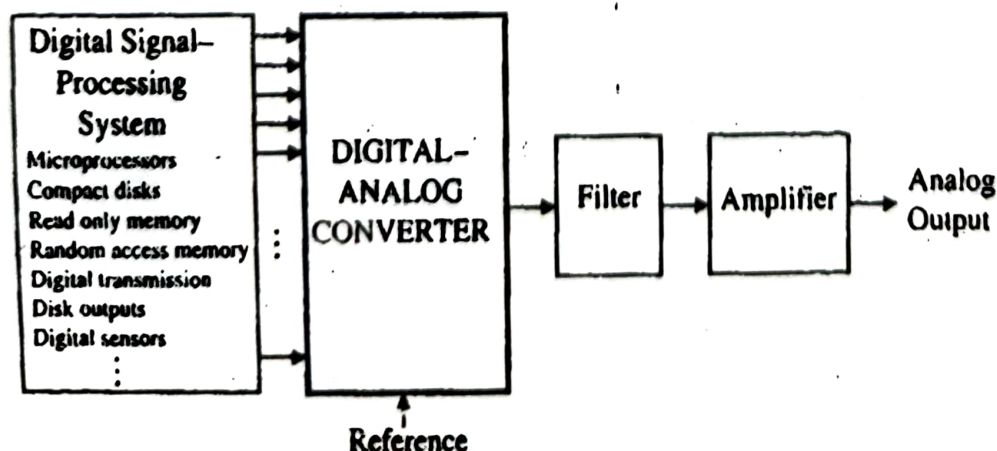


Figure 10.1-1 Digital-analog converter in signal-processing applications.

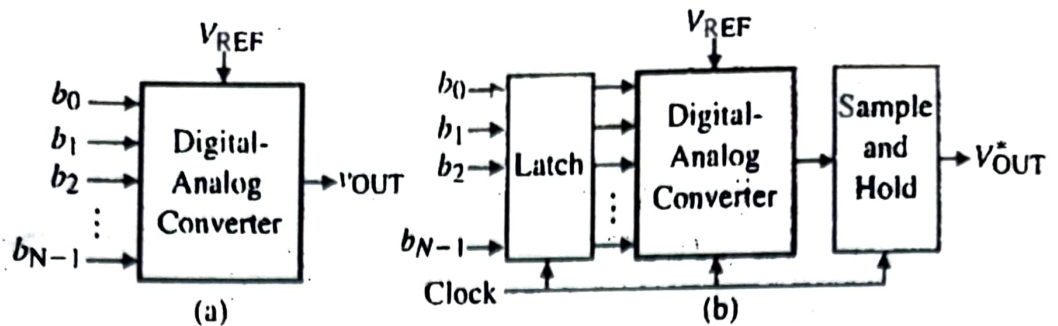


Figure 10.1-2 (a) Digital-analog converter in signal-processing applications. (b) Clocked digital-analog converter for synchronous operation. (The asterisk represents a signal that has been sampled and held.)

In many cases the digital word is synchronously clocked. In this case latches must be used to hold the word for conversion and a sample-and-hold circuit is needed at the output, as shown in Fig. 10.1-2(b). We will examine sample-and-hold circuits in more detail later in this chapter.

The basic form of a DAC providing an analog output voltage is shown in more detail in Fig. 10.1-3. It includes binary switches, a scaling network, and an output amplifier. The scaling network and binary switches operate on the reference voltage to create a voltage that has been scaled by the digital word. The scaling mechanism may be voltage, current, or charge scaling. The output amplifier amplifies the scaled voltage signal to a desired level and provides the ability to source or sink current into a load.

Static Characteristics of DACs

The characterization of DACs is very important in understanding its design. The characteristics of the digital-analog converter can be divided into static and dynamic properties. The resolution of the DAC is equal to the number of bits in the applied digital input word. The resolution of a DAC is expressed as N -bits, where N is the number of bits. Figure 10.1-4 shows the input and output characteristics of an ideal 3-bit DAC ($N = 3$). We see that each of the eight possible digital words has its own unique analog output voltage. These levels are separated by an LSB . The value of the LSB can be defined as,

$$LSB = \frac{V_{REF}}{2^N} \quad (10.1-5)$$

As the digital word increases by 1-bit, the output of the ideal DAC should jump up by $1LSB$. We note that the output is 0.0625 V for the digital input of 000. However, there is no reason

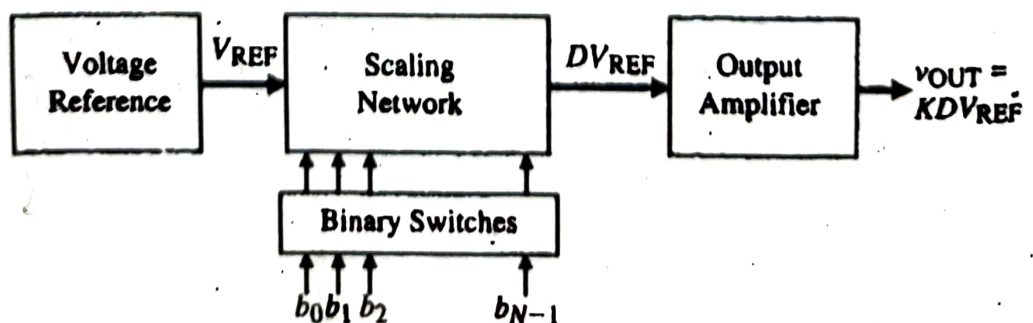
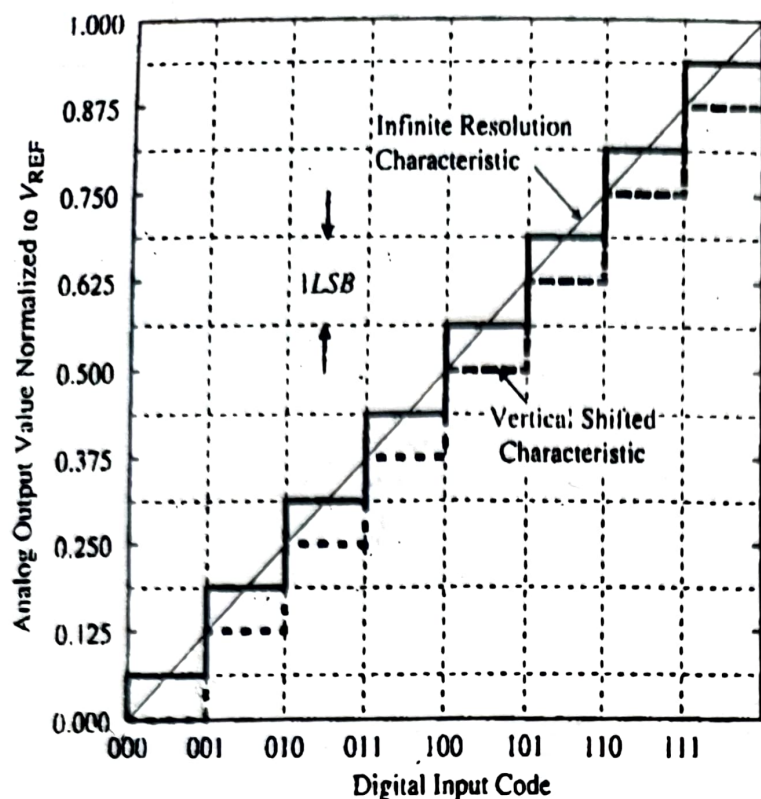


Figure 10.1-3 Block diagram of a digital-analog converter.

Figure 10.1-4 Ideal input-output characteristics of a 3-bit DAC.



why the characteristic cannot be shifted downward by a half LSB as shown by the dashed characteristic, which corresponds to 0 V for the digital input of 000.

Because the resolution of the DAC is finite (3 in the case of Fig. 10.1-4), the maximum analog output voltage does not equal V_{REF} . This result is characterized by the full scale (FS) value of the DAC. The full scale value is defined as the difference between the analog output for the largest digital word (1111 . . .) and the analog output for the smallest digital word (0000 . . .). In general, the full scale of a DAC can be expressed as

$$\text{Full scale (FS)} = V_{REF} - LSB = V_{REF} \left(1 - \frac{1}{2^N} \right) \quad (10.1-6)$$

This definition of FS holds regardless of whether the characteristic has been shifted vertically by $\pm 0.5LSB$. In the case of Fig. 10.1-4, FS is equal to $0.875V_{REF}$. The full scale range (FSR) is defined as

$$FSR = \lim_{N \rightarrow \infty} FS = V_{REF} \quad (10.1-7)$$

Based on the above discussion, let us define several important quantities that are important for DACs. The first is called quantization noise. Quantization noise is the inherent uncertainty in digitizing an analog value with a finite resolution converter. To understand this definition, the characteristic for an infinite resolution DAC is plotted on Fig. 10.1-4. This line represents the limit of the finite DAC characteristic as the number of bits, N , approaches infinity. The quantization noise (or error) is equal to the analog output of the infinite-bit DAC minus the analog output of the finite-bit DAC. If we plot the quantization noise of either of the 3-bit characteristics in Fig. 10.1-4, we obtain the result shown in Fig. 10.1-5. The solid and dashed lines correspond to the solid and dashed staircases, respectively, in Fig. 10.1-4.

2
P.V.
228 229 bit DAC - over a finite bit

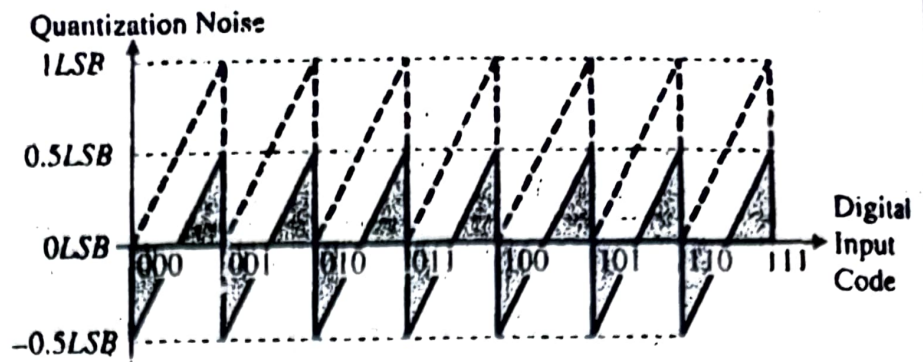


Figure 10.1-5 Quantization noise for the 3-bit DAC of Fig. 10.1-4.

We see from Fig. 10.1-5 that the quantization noise is a sawtooth waveform having a peak-to-peak value of 1LSB . It is useful to note that 0.5LSB is equivalent to $\text{FSR}/2^{N+1}$. This noise is a fundamental property of DACs and represents the limit of accuracy of the converter. For example, it is sufficient to reduce the inaccuracies of the DAC to within $\pm 0.5\text{LSB}$. Any further decrease is masked by the quantization noise that can only be reduced by increasing the resolution.

The dynamic range (DR) of a DAC is the ratio of the FSR to the smallest difference that can be resolved (i.e., an LSB). We can express the dynamic range of the DAC as

$$\text{DR} = \frac{\text{FSR}}{\text{LSB}} = \frac{\text{FSR}}{(\text{FSR}/2^N)} = 2^N \quad (10.1-8)$$

In terms of decibels, Eq. (10.1-8) can be expressed as

$$\text{DR}(\text{dB}) = 6.02N \text{ dB} \quad (10.1-9)$$

The signal-to-noise ratio (SNR) for the DAC is defined as the ratio of the full scale value to the rms value of the quantization noise. The rms value of the quantization noise can be found by taking the root mean square of the quantization noise. For the quantization noise designated by the solid line, this results in

$$\text{rms(quantization noise)} = \sqrt{\frac{1}{T} \int_0^T \text{LSB}^2 \left(\frac{t}{T} - 0.5 \right)^2 dt} = \frac{\text{LSB}}{\sqrt{12}} = \frac{\text{FSR}}{2^N \sqrt{12}} \quad (10.1-10)$$

Therefore, the signal-to-noise ratio of the DAC can be expressed as

$$\text{SNR} = \frac{v_{\text{OUT rms}}}{(\text{FSR}/2^N \sqrt{12})} \quad (10.1-11)$$

The largest possible rms value of v_{OUT} is $(\text{FSR}/2)/\sqrt{2}$ or $V_{\text{REF}}/(2\sqrt{2})$ assuming a sinusoidal waveform. Therefore, the maximum SNR required for a DAC is

$$\text{SNR}_{\text{max}} = \frac{\text{FSR}/(2\sqrt{2})}{\text{FSR}/(2^N \sqrt{12})} = \frac{2^N \sqrt{6}}{2} \quad (10.1-12)$$

In terms of decibels we can rewrite Eq. (10.1-12) as

$$\begin{aligned} \text{SNR}_{\text{max}}(\text{dB}) &= 20 \log_{10} \left(\frac{2^N \sqrt{6}}{2} \right) = 20 \log_{10} (2^N) + 10 \log_{10}(6) - 20 \log_{10}(2) \\ &= 6.02N \text{ dB} + 7.78 \text{ dB} - 6.02 \text{ dB} = 6.02N \text{ dB} + 1.76 \text{ dB} \end{aligned} \quad (10.1-13)$$

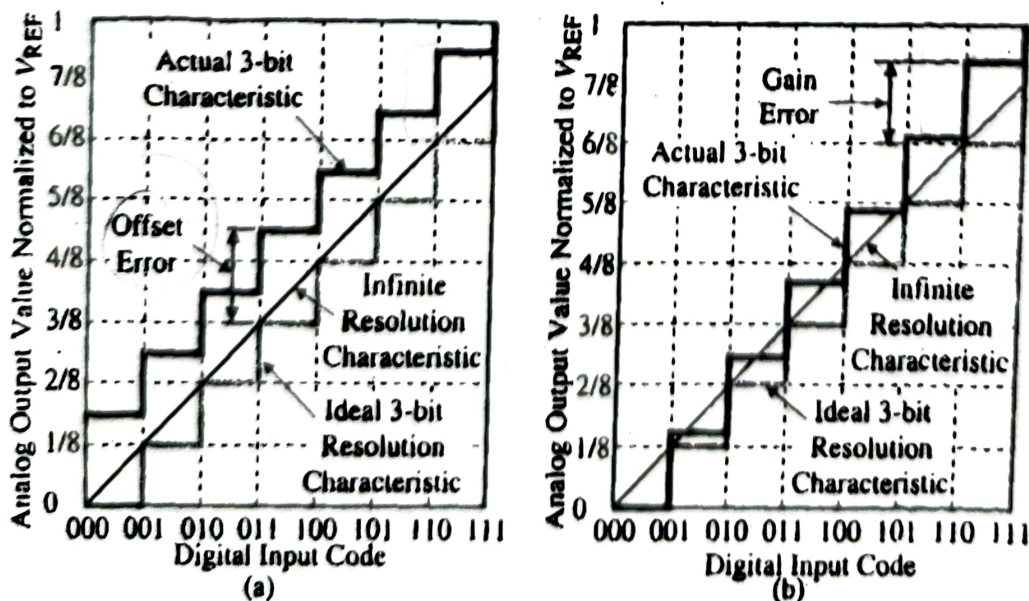
The *effective number of bits (ENOB)* can be defined from the above as

$$\text{ENOB} = \frac{\text{SNR}_{\text{actual}} - 1.76}{6.02} \quad (10.1-14)$$

where $\text{SNR}_{\text{actual}}$ is the actual SNR of the converter.

It is important to summarize the above results because they are very useful in understanding the performance of DACs. The dynamic range necessary for an N -bit DAC is $6N$ dB. The dynamic range can be thought of as the amplitude range necessary to resolve N -bits, regardless of the amplitude of the output voltage. However, when referenced to a given output analog signal amplitude, the dynamic range required must include 1.76 dB more to account for the presence of quantization noise. Thus, for a 10-bit DAC, the DR is 60.2 dB. For a full scale rms output voltage, the signal must be approximately 62 dB above whatever rms voltage level is present in the output of the DAC. This rms voltage could be due to the noise of the op amps and switches or due to the nonlinearity of the op amp.

For each digital word, there should be a unique analog output signal. Any deviations from Fig. 10.1-4 fall into the category of static-conversion errors. Static-conversion errors include offset errors, gain errors, integral nonlinearity, differential nonlinearity, and monotonicity. An offset error is a constant difference between the actual finite resolution characteristic and the ideal finite resolution characteristic measured at any vertical jump. This error is illustrated in Fig. 10.1-6(a). This error could be eliminated by shifting the resulting characteristic vertically. Gain error is the difference between the actual finite resolution and an infinite resolution



Actual
∞ 2 bits
at large
vertical

Figure 10.1-6 (a) Illustration of offset error in a 3-bit DAC. (b) Illustration of gain error in a 3-bit DAC.

characteristic measured at the rightmost vertical jump. Gain error is proportional to the magnitude of the DAC output voltage. This error is illustrated on the 3-bit DAC characteristic shown in Fig. 10.1-6(b).

Integral nonlinearity (INL) is the maximum difference between the actual finite resolution characteristic and the ideal finite resolution characteristic measured vertically. Integral nonlinearity can be expressed as a percentage of the full scale range or in terms of the least significant bit. Integral nonlinearity has several subcategories, which include absolute, best-straight-line, and end-point linearity [2]. The *INL* of a 3-bit DAC characteristic is illustrated in Fig. 10.1-7. The *INL* of an *N*-bit DAC can be expressed as a positive *INL* and a negative *INL*. The positive *INL* is the maximum positive *INL*. The negative *INL* is the maximum negative *INL*. In Fig. 10.1-7, the maximum *+INL* is *1.5LSB* and the maximum *-INL* is *-1.0LSB*.

Differential nonlinearity (DNL) is a measure of the separation between adjacent levels measured at each vertical jump. Differential nonlinearity measures bit-to-bit deviations from ideal output steps, rather than along the entire output range. If V_{cx} is the actual voltage change on a bit-to-bit basis and V_s is the ideal change, then the differential nonlinearity can be expressed as

$$\text{Differential nonlinearity (DNL)} = \left(\frac{V_{cx} - V_s}{V_s} \right) \times 100\% = \left(\frac{V_{cx}}{V_s} - 1 \right) \text{LSBs} \quad (10.1-15)$$

For an *N*-bit DAC and a full scale voltage range of V_{FSR} ,

$$V_s = \frac{V_{FSR}}{2^N} \quad (10.1-16)$$

Figure 10.1-7 also illustrates differential nonlinearity. Note that *DNL* is a measure of the step size and is totally independent of how far the actual step change may be away from the infinite resolution characteristic at the jump. The change from 101 to 110 results in a maximum *+DNL* of *1.5LSBs* ($V_{cx}/V_s = 2.5\text{LSBs}$). The maximum negative *DNL* is found when the digital input code changes from 011 to 100. The change is *-0.5LSB* ($V_{cx}/V_s = -0.5\text{LSB}$), which gives a *DNL* of *-1.5LSBs*. It is of interest to note that as the digital input code changes from 100 to 101, no change occurs (point A). Because we know that a change should have occurred, we can say that the *DNL* at point A is *-1LSB*.

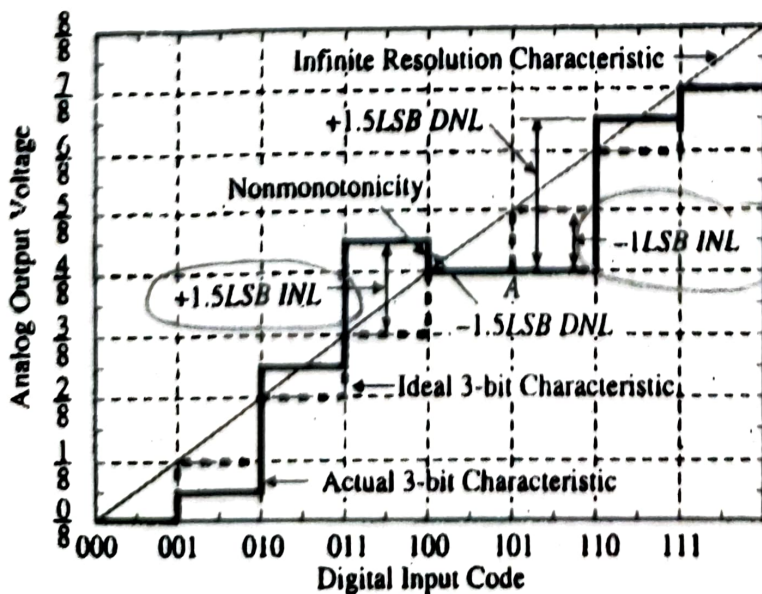


Figure 10.1-7 Illustration of *INL*, *DNL*, and nonmonotonicity in a 3-bit DAC.

Monotonicity in a DAC means that as the digital input to the converter increases over its full scale range, the analog output never exhibits a decrease between one conversion step and the next. In other words, the slope of the transfer characteristic is never negative in a monotonic converter. Figure 10.1-7 exhibited nonmonotonic behavior as the digital input code changed from 011 to 100. Obviously, a nonmonotonic DAC has very poor *DNL*. As a matter of fact, a DAC that has a $-DNL$ that is $-1LSB$ or more negative will always be nonmonotonic.